Roll Number: _____ Name: _____

Thapar Institute of Engineering & Technology, Patiala
(Deemed to be University)
Department of Electrical and Instrumentation Engineering
Auxiliary Exam

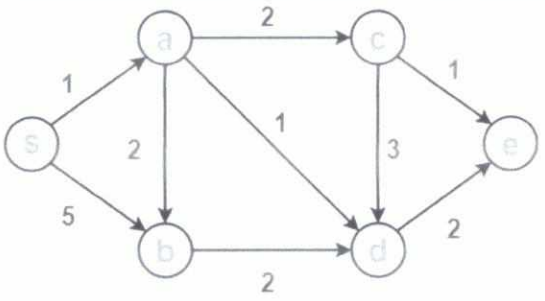
Max. Marks: 100
Time: 03 Hours
Date: 27-08-2025 (17:30-20:30)

Course Name: Computer Networks (UCS520)

Name of Faculty: Dr. Aditya Gupta

Note: * Attempt any five questions. Assume missing data, if any, suitably. Attempt all parts of a question at one place.

Q1	(a) What are the differences between TCP/IP Model and OSI Model?	**[7]**
	(b) Differentiate between physical topology and logical topology.	**[7]**
	(c) What do you mean by message switched networks?	**[6]**
Q2	(a) Consider the binary code that consists of only four valid codewords as given below: 00000, 01011, 10101, 11110 Let the minimum Hamming distance of the code p and the maximum number of erroneous bits that can be corrected by the code be q. Then find the values of p and q.	**[10]**
	(b) Explain the concept of minimum Hamming distance (d_min) and show how it determines the error detection and correction capability of a code.	**[5]**
	(c) Given 1101011011 data frame and generator polynomial $G(x) = x^4 + x + 1$. Derive the transmitted frame.	**[5]**
Q3	(a) Given a network address of 192.18.100.0 and a subnet mask of 255.255.255.192. i) How many subnets are created?	**[10]**
	ii)How many hosts are there per subnet?	**[5]**
	(b) What are the differences between IPv4 and IPv6 Addressing?	
	(c) What role does NAT (Network Address Translation) play in IP addressing?	**[5]**
Q4	(a) Explain about Dijkstra shortest path algorithm?	**[6]**
	(b) Distinguish ARP and RARP protocols and their services	**[4]**
	(c) Apply the Dijkstra shortest path algorithm to the following graph to find the shortest path.	**[10]**

		
Q5	(a) A server application is running on port 80 and another on port 21. Explain how the Transport Layer ensures that data is delivered to the correct application. (b) Describe the TCP three-way handshake process with a diagram. What purpose does it serve?	**[10]** **[10]**
Q6	(a) Write a brief note on the following i) World Wide Web ii) E-Mail iii) Gateways iv) TELNET (b) Consider a scenario where a client uploads a file to a remote server. Explain step-by-step how FTP enables this process, including authentication and data transfer modes.	**[10]** **[10]**

********** All the Best **********